

Integrazione e Immigrazione come politiche per la natalità

Integration and immigration as policies to boost birth rates

Demographic Projections and Migration Policy for Italy

Final report

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1. Our Project, Scenarios, and Research Question

This project studies how the transformation of Italy's age structure affects births and evaluates whether immigration and integration can work as complementary demographic policy tools. The starting point is that Italy's demographic decline cannot be understood only through the total fertility rate. TFR measures fertility intensity, but the number of births also depends on the number and age distribution of potential parents. When the cohorts entering reproductive ages are small, even a recovery in fertility may produce only a limited increase in births. For this reason, the project focuses on the structural side of the demographic process: the size and composition of the population by age. The empirical core of the project is a cohort-component population projection for Italy under alternative migration scenarios.

The project has three core aims. First, it reconstructs the evolution of the number of potential parents over time by age and sex. Second, it builds alternative migration scenarios that differ in both the total number of immigrants and the age profile of inflows. Third, it discusses policy levers that can increase the number of potential parents, combining immigration levels, selective age profiles, family composition, and stronger social and civic integration.

The project works with three main migration scenarios and one reverse-policy calculation:

Scenario	Definition in the project
Scenario 1: baseline	Migration continues according to the internally forecast baseline path for immigration and emigration, using the current/recent age-sex-origin structure as the reference.
Scenario 2: policy on levels	The total number of immigrants is increased, while the baseline age-sex-origin structure is kept fixed.
Scenario 3: targeted policy	The total number of immigrants is increased, and the age structure is shifted toward younger adult ages and children, representing a policy oriented to potential parents and families.
Reverse policy calculation	The tool can also solve for how many additional immigrants per year are needed to reach a selected population or OADR target. This is used in addition to forward scenario simulation, not instead of it.

Table 1: Our Scenarios

The research question is: under the two policy-oriented scenarios, how many immigrants would be required to reduce depopulation pressure and keep the old-age dependency ratio below a chosen threshold? Operationally, we treat this also as an inverse problem: in addition to fixing migration assumptions and observing outcomes, the tool can solve for the migration level and profile required to approach defined targets.

2. State of the Art: What ISTAT and UN Do

2.1 ISTAT

ISTAT population projections are produced using the cohort-component method, where the future population is obtained by projecting fertility, mortality and international migration and applying them to the population by age and sex.

Fertility

Regional fertility projections are based on fertility intensity and age schedule of fertility.

The total fertility rate (TFR) is forecast using ARIMA time-series models estimated on historical data (1977–2021). The model most frequently used is ARIMA (2,0,0) with intercept, estimated separately for each region.

The age pattern of fertility is modelled using the quadratic spline model proposed by Schmertmann. This approach represents the standardized age-specific fertility curve through three parameters:

- α : age at the beginning of the reproductive schedule;
- P : age at peak fertility;
- H : age after the peak at which fertility declines to half of the maximum level.

The quadratic spline model fits five second-degree polynomials to the fertility schedule. The resulting function is continuous with a continuous first derivative and is uniquely determined by the parameter vector $[\alpha, P, H]$.

Forecasting age-specific fertility rates therefore reduces to forecasting the three parameters of the model using ARIMA processes estimated on historical data.

Mortality

Mortality projections are produced using the Lee-Carter model in the Lee–Miller variant. In this formulation the logarithm of the age-specific death probabilities is approximated using three parameters:

$$\ln(q_x(t)) = a(x) + b(x)k(t)$$

where

- $a(x)$ represents the average age pattern of mortality;
- $b(x)$ represents the sensitivity of mortality at age x to changes in the mortality index;
- $k(t)$ represents the overall mortality trend over time.

In the Lee–Miller version used by ISTAT, the adjustment procedure ensures that the fitted mortality schedule reproduces the observed life expectancy at birth, rather than the total number of deaths as in the original Lee–Carter formulation.

Future mortality levels are obtained by projecting the time index $k(t)$, estimated on the period 1974–2021, using a random walk with drift. The parameters $a(x)$ and $b(x)$ are initially kept constant over time.

International migration

Projections of international migration flows are constructed using a deterministic intermediate model based on recent observations.

The total number of immigrants and emigrants in the first projection year is set equal to the average observed during the most recent five valid years (2016–2019 and 2021). Over the projection horizon, immigration and emigration totals are assumed to converge gradually to the same level, corresponding to the initial semi-sum of the two flows.

The age-sex distribution of migration flows is obtained using the Castro–Rogers model. This model represents the typical age profile of migration as a mathematical function composed of several additive components and up to eleven parameters. These parameters are estimated using the observed migration flows in 2016–2019 and 2021 through nonlinear regression and are kept constant during the projection period, implying that overall migration level may change over time while the age distribution remains fixed.

2.2 United Nations

The UN produces projections with a cohort-component framework using single years of age and annual time steps, starting from 1 January 2024 and projecting forward to 2101. The UN uses probabilistic methods to generate many trajectories for the key drivers: TFR, life expectancy at birth, and total net migration.

Fertility

TFR is projected probabilistically using a Bayesian hierarchical model built around a double-logistic representation of fertility decline. In practice, the UN generates large samples of double-logistic curves (via MCMC), then produces a set of projected TFR trajectories to represent uncertainty. Age-specific fertility rates are then derived in a way that is consistent with the projected TFR path.

Mortality

The UN projects life expectancy at birth by sex using probabilistic models. In WPP2024, female life expectancy is projected with a model of the mortality transition where gains in life expectancy depend

on the current level of life expectancy (Raftery et al., 2013), estimated in a Bayesian hierarchical setting.

Once the future path of life expectancy is set, the UN derives age-specific mortality rates by single year of age and sex using different methods depending on data quality.

Migration

A key difference vs ISTAT is that the UN incorporates international migration as net migration only.

For WPP2024, total net migration is projected probabilistically using methods developed by Azose & Raftery, i.e., a Bayesian hierarchical approach for future trajectories of the crude net migration rate and net migration counts.

For the age-sex distribution, WPP estimates or assigns patterns based on available evidence. In WPP2024, a new approach is introduced: total net migration is first split into implied immigration, and emigration counts and then a Rogers–Castro model is used to apportion by age and sex.

3. Our Methods

We implement a cohort-component projection pipeline that follows the broad logic used by official statistical agencies, while keeping the model transparent and adaptable for policy discussion. The system is designed to produce population projections by single year of age, sex, and origin/citizenship, together with fertility, mortality, migration, and uncertainty outputs.

Component	Implemented output
Mortality	Age-sex survival probabilities and life expectancy at birth by sex.
Fertility	Age-specific fertility rates, total fertility rate, and births by sex.
Immigration	Baseline immigration forecast by age, sex, and origin, plus scenario modifiers.
Emigration	Baseline emigration forecast by age, sex, and origin.
Population projection	Annual cohort-component update by age, sex, and origin/citizenship.
Uncertainty	Simulation paths summarized through medians and prediction intervals.

[2: Our Methods](#)

3.1 Mortality

Mortality is forecast by age and sex. The implemented approach follows the logic of Lee-Carter and Lee-Miller models: historical age-specific mortality is decomposed into an average age pattern, an age-specific sensitivity, and a time trend. The future mortality path is then generated by projecting the time component forward and reconstructing age-sex survival probabilities.

The Streamlit tool allows the user to exclude COVID years from the mortality fit. This is important because the pandemic years can distort the structural mortality trend if treated as ordinary

observations. The output of the mortality module is a set of age-sex survival probabilities and life expectancy at birth by sex, which enter directly into the cohort-component projection.

3.2 Fertility

Fertility is modelled through both intensity and age schedule. The intensity component is summarized by the total fertility rate, while the age schedule is represented by age-specific fertility rates. This follows the logic discussed in the original report: fertility projections require both a level and an age pattern.

The implemented system allows the user to modify TFR assumptions separately for Italians and foreigners. These changes can be applied immediately across the forecast horizon or smoothly toward the final year. Births are then calculated by multiplying the female population at reproductive ages by the projected age-specific fertility rates. Newborns are split by sex using the sex ratio at birth.

3.3 Emigration vs Immigration

The final implementation explicitly separates immigration and emigration. Both are forecast as baseline demographic components, and both are then allocated by age, sex, and origin/citizenship. This corrects the earlier draft formulation in which immigration was described mainly as a scenario variable.

Emigration is treated as a forecasted component. Historical emigration totals and age profiles are used to build a baseline path, and broad observed age ranges are expanded into single ages through smoothed age profiles. The output is emigration by age, sex, origin/citizenship, and year.

Immigration is also forecast in the baseline, because a demographic baseline without immigration would not represent the current functioning of the Italian population system. However, immigration is also the main policy lever. The policy scenarios therefore modify baseline immigration after the baseline has been produced. Scenario 2 increases total immigration while preserving the baseline age-sex-origin structure. Scenario 3 increases immigration and reshapes the additional inflow toward younger adult ages and children.

The age distribution of migration is reconstructed from observed broad age ranges. The code expands these age ranges to single ages using smoothed templates informed by the current age-sex structure of Italians and foreigners, while rescaling inside each broad age band so that the original observed age-band totals are preserved.

3.4 Uncertainty: Simulation-Based Projection (ISTAT-Style)

Uncertainty is propagated through simulation paths. Each simulation corresponds to a coherent set of fertility, mortality, and migration inputs. These inputs are passed through the same cohort-component engine, producing one complete future population trajectory.

Following the logic of ISTAT-style probabilistic projection, the model summarizes the resulting trajectories through median paths and prediction intervals. In the Streamlit version, the number of simulations is user-controlled and kept moderate by default so the app remains fast enough for live use. Larger simulation runs can be used outside the app when publication-quality uncertainty intervals are required.



Figure 1: Methodology framework

4. Data: ISTAT and UN Inputs, and What They Provide

This section clarifies which inputs are used for the projection pipeline and what each external source contributes. ISTAT is the main source for constructing inputs that can be split by age, sex, and, where

available, citizenship/origin. UN WPP is used mainly as an external benchmark for aggregate trends and uncertainty framing.

4.1 ISTAT

ISTAT provides the key national demographic inputs used to build the projection system: baseline population by single year of age and sex, population split by origin/citizenship, fertility data including births and age-specific fertility patterns, mortality by age and sex, and observed immigration and emigration flows.

Fertility rates by mother's age, event year and mother's citizenship (per thousand women)

Frequency: Annual, Indicator: Age-specific fertility rate

Criteria

Pivoting

Table

Chart

Map

Territory

Italy

Mother's age	Until 14 years	15 years	16 years	17 years	18 years	19 years	20 years	21 years	22 years	23 years	24 years	25 years
Time												
Citizenship: Italian												
2002	0	0,12	2,94	5,57	8,91	12,58	16,3	20,95	26,64	33,82	41,33	
2003	0,03	0,14	2,99	5,37	8,48	12,44	16,03	20,12	25,67	32,43	40,69	
2004	0,02	0,15	2,92	5,44	8,31	12,3	16,02	20,31	25,58	31,32	39,44	
2005	0	0,12	2,82	5,5	8,22	12,25	15,69	19,77	24,27	30,2	37,78	
2006	0,03	0,12	2,6	4,82	7,99	12,14	15,47	18,83	23,69	30,23	37,2	
2007	0,01	0,13	2,62	4,74	7,84	11,72	15,64	18,98	23,6	28,74	36,55	
2008	0,02	0,18	2,83	5,07	8,13	11,56	15,28	19,3	24	30,18	37,1	

Figure 2: Istat dataset on fertility

Singole età

Frequency: Annuale

Criteria	Territorio	Sesso	Tempo	Indicatore	Sopravviventi - lx	Decessi - dx	Probabilità di morte (per 1.000) - qx	Anni vissuti - Lx	Probabilità prospettiva di sopravvivenza - Px	Speranza di vita - ex
Pivoting	Italia	Totale	2024	Età						
Metadati Referenziali				0 anni	100.000	252	2,52262	99.763	0,9997495	83,519
Tabella				1 anni	99.748	20	0,20267	99.738	0,999828	82,729
Grafico				2 anni	99.728	14	0,14125	99.720	0,9998765	81,746
Mappa				3 anni	99.713	11	0,10583	99.708	0,9999034	80,757
				4 anni	99.703	9	0,08742	99.699	0,9999136	79,766
				5 anni	99.694	9	0,08544	99.690	0,9999147	78,773
				6 anni	99.686	8	0,0851	99.681	0,9999173	77,779
				7 anni	99.677	8	0,08027	99.673	0,9999214	76,786
				8 anni	99.669	8	0,07702	99.665	0,9999246	75,792
				9 anni	99.661	7	0,07374	99.658	0,999927	74,798
				10 anni	99.654	7	0,07233	99.651	0,9999264	73,803

Figure 3: Istat dataset on mortality

Immigrazione e cittadinanza

Frequency: Annuale, Territorio di origine: Totale, Indicatore: Iscritti in anagrafe, Tipo di trasferimento: Estero

Criteria	Tempo	Sesso	Cittadinanza	Maschi	Femmine	Totale
Selezioni	2024		Età: Fino a 17 anni			
Personalizza visualizzazione			Francia	177	173	350
Metadati Referenziali			Germania	167	163	330
Tabella			Grecia	59	56	115
Grafico			Irlanda	17	20	37
			Italia	6660	6074	12.734
			Kosovo	367	257	624
			Lettonia	24	15	39

Figure 4: Istat dataset on migrations

Where the input files contain citizenship or origin distinctions, the model preserves them and uses them in the cohort-component projection.

For migration, the input files contain observed migration counts by broad age range. These are reconstructed into single ages using the age-profile logic described above. This allows the projection to operate at the single-age level while remaining consistent with the observed broad age-band totals.

4.2 UN Data (World Population Prospects)

UN WPP data are used as external comparisons rather than as the main engine of the projection. The relevant UN information includes projected population totals, fertility trajectories, life expectancy paths, and net migration trajectories. In the Streamlit app, UN comparison data can be displayed alongside our projection to help users evaluate whether the internally generated paths are plausible relative to a major international benchmark.

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Migration

Excel files with annual estimates and standard projection scenarios (see [Definition of Projection Scenarios](#) for further details). For probabilistic results, select [Probabilistic Projections](#) above. For advanced users who need to use these data in a database form or statistical software, we recommend to use the [CSV format](#) for bulk download.

Sub Group	Files (Click to Download)	Description
Probabilistic	Number of Births (XLSX)	Total births (both sexes combined). Data are presented in thousands.
	Crude Birth Rate (XLSX)	Crude birth rate (births per 1,000 population).
	Total Fertility (XLSX)	Total fertility (live births per woman).
	Age specific fertility rate - 5 years (XLSX)	Age specific fertility rate - 5 years.

Figure 5: UN WPP Database

			Age-specific fertility rates (births per 1,000 women)											
Index	Variant	Region, subregion, country or area *	Year	10	10-14	11	12	13	14	15	15-19	16	17	18
14559	Median PI	Italy	2025	0,0	0,0	0,0	0,0	0,0	0,0	0,3	2,9	0,8	2,1	4,2
14560	Median PI	Italy	2026	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,0	0,8	2,1	4,2
14561	Median PI	Italy	2027	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,0	0,8	2,1	4,2
14562	Median PI	Italy	2028	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,0	0,8	2,1	4,3
14563	Median PI	Italy	2029	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,1	0,8	2,1	4,3
14564	Median PI	Italy	2030	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,1	0,8	2,1	4,3
14565	Median PI	Italy	2031	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,1	0,8	2,1	4,3
14566	Median PI	Italy	2032	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,1	0,8	2,1	4,3
14567	Median PI	Italy	2033	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,1	0,8	2,2	4,4
14568	Median PI	Italy	2034	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14569	Median PI	Italy	2035	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14570	Median PI	Italy	2036	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14571	Median PI	Italy	2037	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14572	Median PI	Italy	2038	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14573	Median PI	Italy	2039	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14574	Median PI	Italy	2040	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14575	Median PI	Italy	2041	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14576	Median PI	Italy	2042	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,8	2,2	4,4
14577	Median PI	Italy	2043	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,9	2,2	4,4
14578	Median PI	Italy	2044	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,9	2,3	4,4
14579	Median PI	Italy	2045	0,0	0,0	0,0	0,0	0,0	0,0	0,3	3,2	0,9	2,3	4,4

Figure 6: UN dataset on forecasted fertility

5. Results

The results of the project are presented primarily through the Streamlit application rather than through a fixed set of static figures. This choice is intentional. The central object of the project is not a single population forecast, but a policy simulator: the relevant question is how Italy's demographic trajectory changes when assumptions on fertility, mortality, immigration levels, and the age structure of migrants are modified.

The application produces a set of core demographic outputs. These include projected total population with uncertainty bands, population pyramids for selected years, annual births, the old-age dependency ratio, total fertility rate, life expectancy at birth by sex, immigration and emigration flows, net migration, and reverse-policy outputs showing the additional annual immigration required to reach a selected population or OADR target.

A first key result concerns the structural nature of Italy's demographic decline. The baseline projection confirms the demographic pressure created by low fertility and population ageing. In addition, the simulator includes an intentionally optimistic fertility stress-test scenario, where the projected TFR rises to values that exceed the upper uncertainty bounds of the official comparison sources. This scenario should not be interpreted as our most credible forecast. Rather, it is included to test whether a very favorable fertility trajectory, taken alone, would be enough to prevent population decline. The result is informative: even under this deliberately optimistic assumption, total population still declines over the projection horizon.

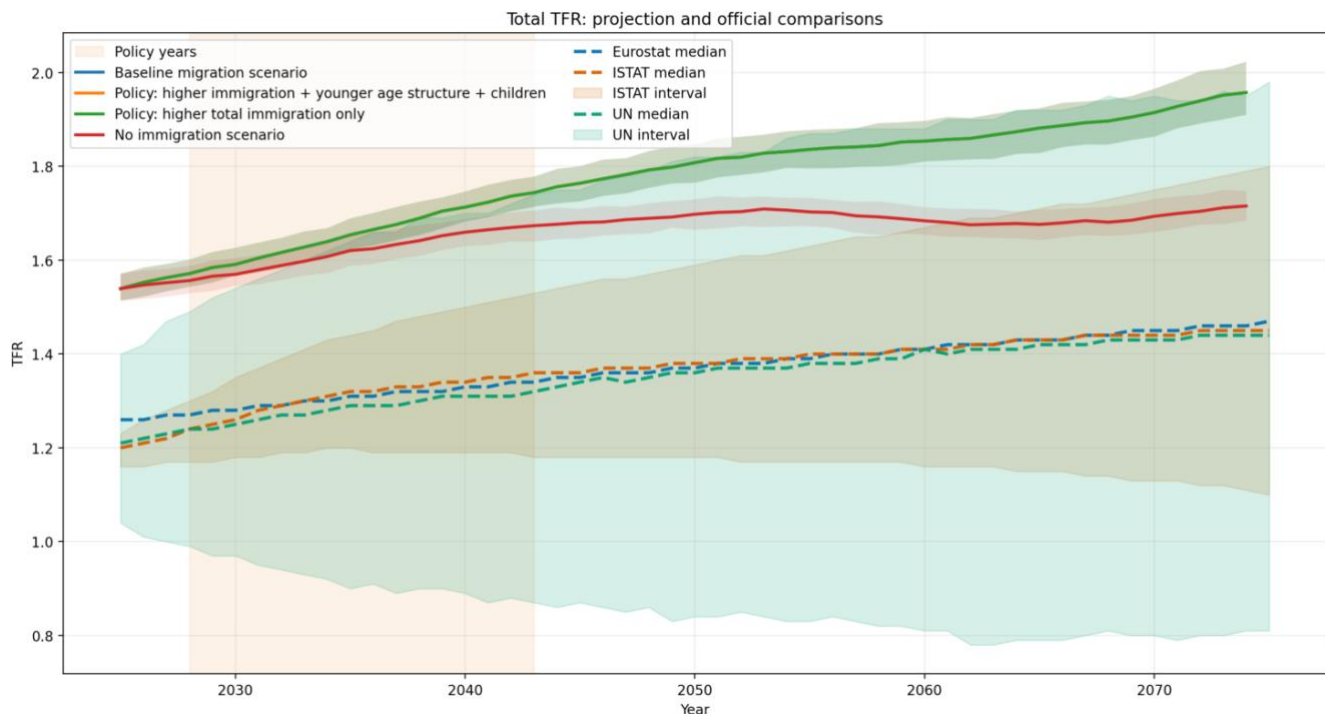


Figure 7: TFR under unlikely high scenario

This supports one of the main points of the project: Italy's demographic challenge is not only linked to how many children each woman has, but also to how many potential parents there will be in the population. Given the current age structure, fertility recovery alone is unlikely to fully offset depopulation.

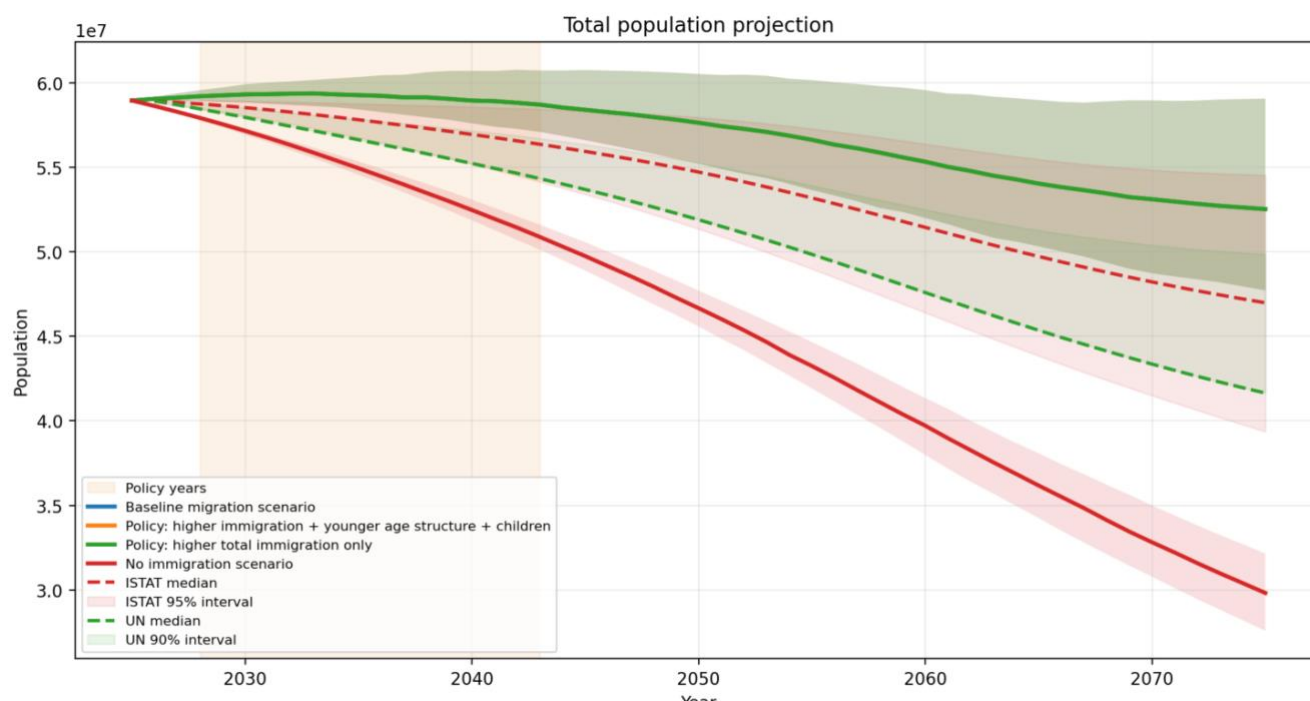


Figure 8: Resulting population under unlikely high TFR

The no-immigration counterfactual is also informative. It is not intended as a realistic policy scenario, but as a diagnostic benchmark. By removing immigration while keeping the other demographic components active, the model shows how strongly future population size and age structure depend on migration. The resulting trajectory highlights the role of migration in slowing depopulation and limiting the worsening of dependency ratios.

The policy scenarios show that the number of immigrants is not the only relevant parameter. The age structure of inflows matters. Additional immigrants concentrated at younger adult ages, especially when combined with children and family migration, have stronger long-term demographic effects than the same number of immigrants distributed according to the baseline age profile. This occurs because younger migrants contribute not only to total population size, but also to the working-age population and, indirectly, to future births.

This point is especially important for the old-age dependency ratio. Keeping the OADR low is not only a matter of increasing the total number of immigrants. It also requires policies that take into account the age composition of inflows. A migration scenario that adds population without considering age may help sustain total numbers, but it can be less effective in improving the balance between working-age and older-age groups. By contrast, inflows concentrated in young adult ages have a stronger effect on the denominator of the OADR and can therefore slow the increase in dependency

more effectively. This suggests that, for ageing-related policy goals, migration policy should be evaluated not only by volume, but also by demographic composition.

The reverse-policy tool provides the most direct policy interpretation. Instead of asking only “what happens if immigration increases?”, the application allows the user to ask “how much additional immigration would be needed to reach a specific demographic target?” This turns the projection into a decision-support instrument: targets such as a desired population size or a maximum OADR can be translated into annual additional migration requirements under alternative assumptions about migrant age structure.

Scenario	Target year & OADR	Required + imm x year	Total + imm policy	Policy years	Status
Policy: higher total immigration only	2075 & 68	x	30000000	15	Not reached below 2,000,000 extra immigrants/year
Policy: higher immigration + younger age structure + children	2075 & 68	91171	1367567	15	Solved

Table 3: Example of additional annual immigration required to reach a selected target

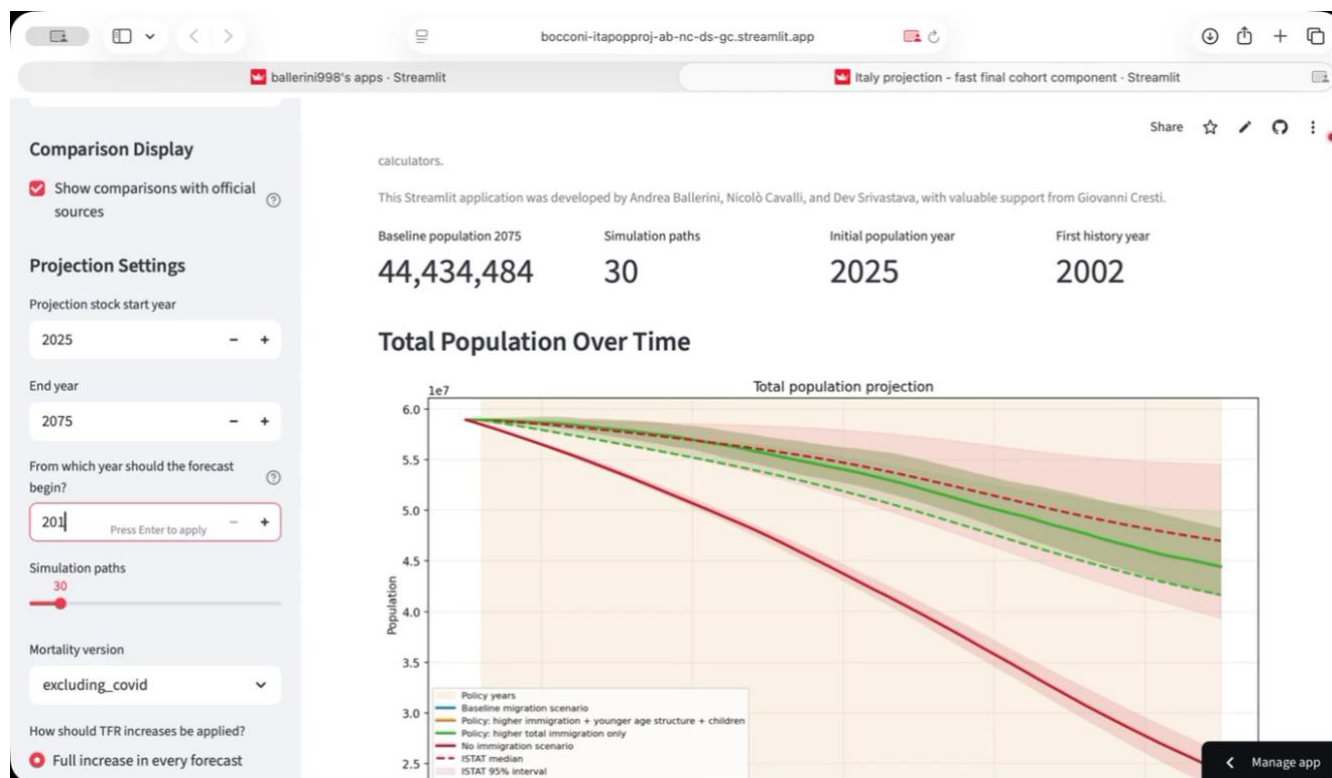
Overall, the results suggest that immigration policy can substantially affect Italy’s demographic trajectory, but its impact depends on timing, scale, and composition. Fertility recovery remains important, but the projections show that even very favorable fertility assumptions are unlikely to be sufficient on their own, given Italy’s current age structure. Policies focused only on aggregate migration numbers are therefore less informative than policies that also consider age, family structure, integration, and long-term residence. For this reason, the Streamlit application is used as the main results interface: it allows stakeholders to explore these mechanisms directly and transparently.

6. Streamlit

The Streamlit application is the operational interface of the project. It allows the user to select the forecast start year, choose the mortality fitting option, modify TFR assumptions separately for Italians and foreigners, define policy years, set the number of additional immigrants, select the target age range for policy migrants, choose the child share of additional migrants, toggle official comparisons, and download CSV outputs.

The app also includes the reverse-policy game. The user can enter a desired total population or OADR in a selected target year, and the tool returns the annual additional immigration required under the two policy scenarios. This makes the policy question concrete and transparent.

The application can be accessed via the following link: <https://bocconi-itapopproj-ab-nc-ds-gc.streamlit.app/>, and a demonstration video of its functionality is available here: https://drive.google.com/file/d/1G5wsrnuSGMd7JF9EUo9lc4ir7QURK0Pk/view?usp=drive_link



Video 1: Streamlit Application

7. Spin-Offs (TBD)

The demographic projection can be used as the base for economic and social extensions. On the economic side, projected age structures and dependency ratios can feed simulations of labour supply, pension expenditure, health expenditure, education demand, and tax-base dynamics under alternative migration policies.

On the social side, projections can be linked to integration and citizenship policy scenarios. This matters because immigration is not only an inflow but also a long-term process involving residence duration, naturalization, family formation, labour-market integration, school participation, and social inclusion.

A major future development is to obtain more granular data by citizenship, country of birth, legal status, duration of stay, and naturalization pathway. With these data, the model could distinguish groups with different fertility patterns, emigration risks, labour-market profiles, and integration trajectories. This would make the economic and social spin-offs substantially more informative for policy.

A further development is regionalization. National projections are useful for strategic discussion, but many policy questions are territorial: schools, housing, labour-market matching, health services, and integration capacity vary strongly across regions and municipalities.

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